Netflix Data Analysis and Storage: A Reflection Using the Netflix Dataset

Dataset overview

For my analysis, I used the Netflix dataset (United States, 2024), which contains 8807 rows and 12 columns, with both qualitative and quantitative variables. After a dataset cleanup, 9 columns remained, including key variables such as type, rating, duration_time, and release_year.

I loaded the dataset locally because it is small, flat, and static.

```
=== Clean dataset ===

(8807, 9)
Index(['show_id', 'type', 'title', 'country', 'date_added', 'release_year',
       'rating', 'duration', 'duration_time'],
      dtype='object')
     show_id     type                                title         country date_added  release_year rating  duration  duration_time
4336   s4337    Movie                 Crossroads: One Two Jaga        Malaysia 2018-12-01          2018  TV-14    81 min           81.0
5726   s5727    Movie  A Grand Night In: The Story of Aardman  United Kingdom 2016-11-01          2015  TV-PG    59 min           59.0
4021   s4022    Movie                            Seven Sundays     Philippines 2019-03-14          2017  TV-MA   128 min          128.0
1560   s1561    Movie                                 The Prom   United States 2020-12-11          2020  PG-13   132 min          132.0
3732   s3733    Movie                    The Edge of Democracy          Brazil 2019-06-19          2019  TV-14   122 min          122.0
1809   s1810    Movie                          The Hows of Us     Philippines 2020-10-22          2018  TV-14   120 min          120.0
7991   s7992  TV Show                          Shadow of Truth          Israel 2017-01-27          2016  TV-MA  1 Season            1.0
4096   s4097    Movie                                 Studio 54   United States 2019-02-16          2018  TV-MA    99 min           99.0
5316   s5317    Movie                       Buster's Mal Heart   United States 2017-08-26          2016  TV-MA    98 min           98.0
144     s145    Movie                              House Party   United States 2021-09-01          1990      R   104 min          104.0
```

Figure 1. Netflix data set sample cleaned

Local versus Cloud Storage

Local data storage enables fast access with minimal overhead, ideal for straightforward analysis of structured data. In contrast, cloud storage offers scalability, improved performance as data grows, shared access, faster recovery, and easier integration with automated pipelines. The ultimate storage choice will depend on organisational needs, the complexity of the analysis to be performed, the data structure, and its growth (Golec, Strugar and Belak, 2021).

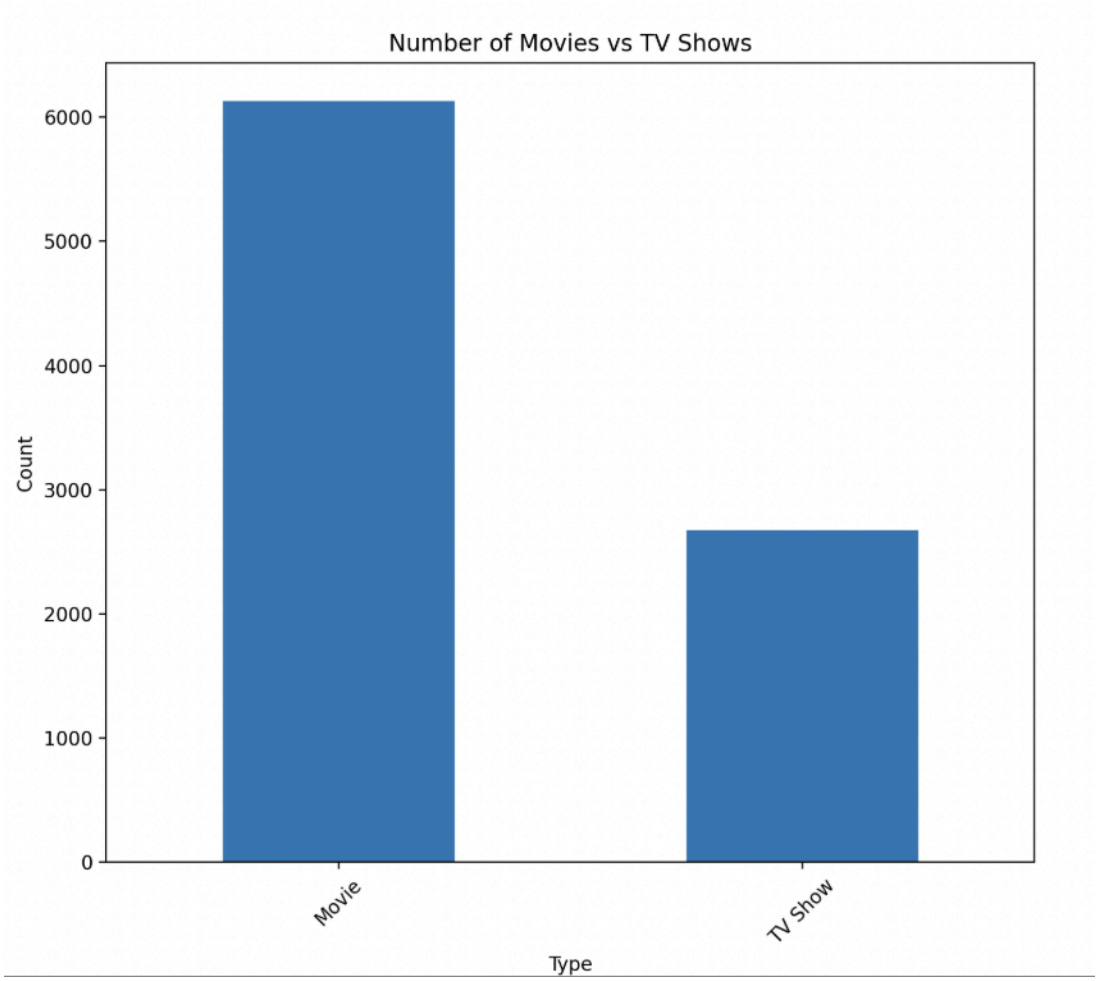
Netflix dataset and Data storage

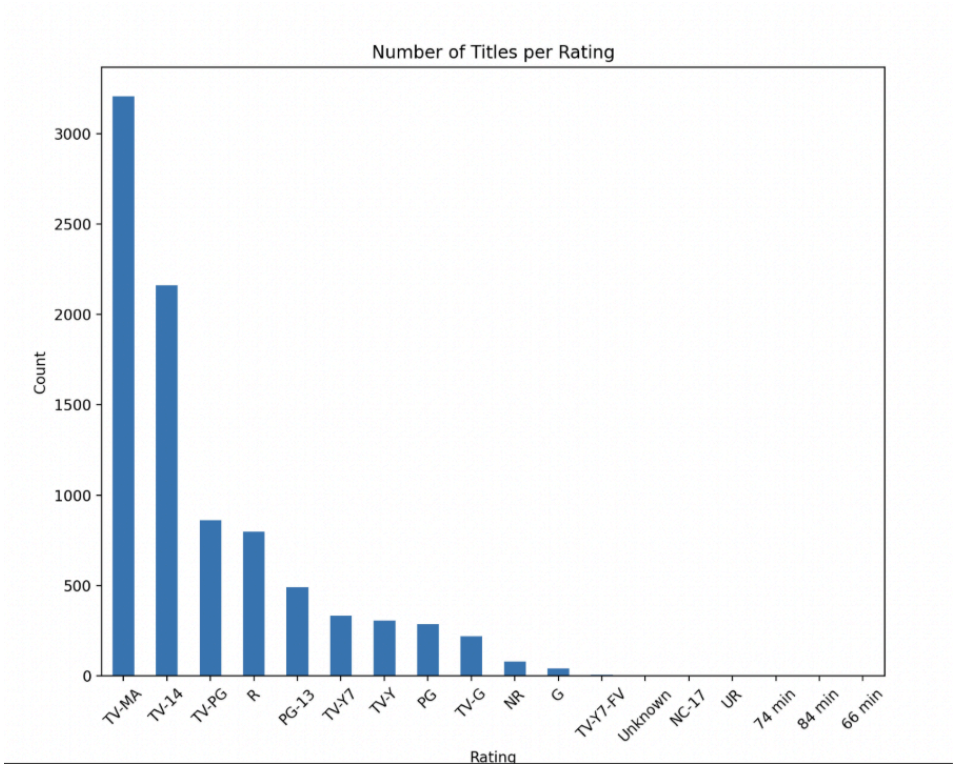
Since my analysis used a simple CSV, basic cleaning, and visualisation in local Python, local storage was optimal for this single-user case. The small, static dataset does not justify the complexity or cost of cloud storage.

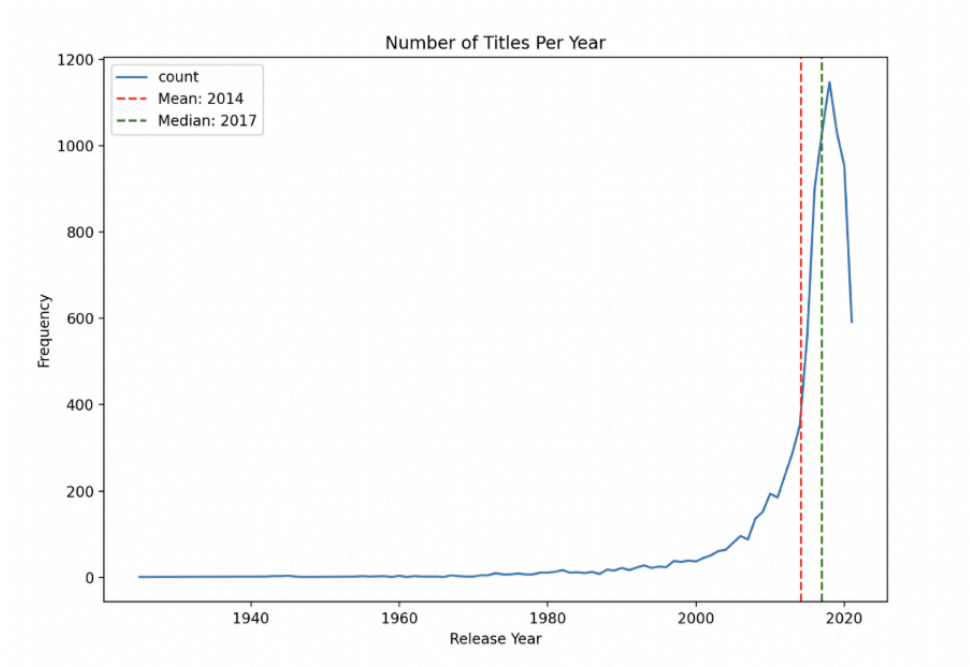
Conclusion

For this straightforward analysis, local storage is the most suitable choice. Cloud storage should be considered only if the dataset becomes larger, dynamic, or part of a collaborative project.

Netflix Dataset Analysis (Python)




Number of Titles per Rating
Count
3000
2500
2000
1500
1000
500
0
TV-MA
TV-14
TV-PG
R
PG-13
TV-Y7
TV-Y
PG
TV-G
NR
G
TV-Y7-FV
Unknown
NC-17
UR
74 min
84 min
66 min
Rating


Number of Titles Per Year
count
Mean: 2014
Median: 2017
Frequency
1200
1000
800
600
400
200
0
1940
1960
1980
2000
2020
Release Year

References

Golec, D., Strugar, I. and Belak, D. (2021) ‘The benefits of enterprise data warehouse implementation in cloud vs. on-premises’, *ENTRENOVA – ENTerprise REsearch InNOVAtion*, 7(1), pp. 66–74. DOI: 10.54820/DMZS9230

Netflix (United States) (2024) ‘Netflix Title Dataset’. *Zenodo*. DOI: 10.5281/zenodo.13925131